

Appendix A - Completed Research Plan

1. Overview

The completed research plan represents the actual execution of the dissertation. The plan reflects the natural progression of tasks across the study, incorporating adjustments made during preprocessing, feature engineering, model development, interpretation, and dissertation writing. It is presented alongside the original planned timeline included in the research proposal, without altering the original document.

2. Completed Research Plan

The completed research plan represents the actual execution of the dissertation. During implementation, several stages of the workflow evolved from their original planned structure to accommodate methodological refinements and iterative analytical requirements. In particular, the literature review extended into the topic confirmation phase, feature engineering and model development required multiple iterative refinements, and dissertation writing progressed in parallel with the later analytical stages. The completed timeline reflects these adjustments while remaining consistent with the original research objectives.

Where the original Gantt chart included overlapping durations for literature review and methodology preparation, the completed plan consolidates these activities into a merged phase representing the actual progression. Similarly, the completed plan includes additional stages such as hyperparameter tuning, validation-based model selection, final test evaluation, and SHAP interpretation, which were introduced as part of methodological refinements during the dissertation. Contingency buffers originally allocated to feature engineering were expanded to reflect the multiple iteration cycles that occurred during preprocessing, feature engineering, and model tuning.

Dissertation writing was executed as an overlapping activity across the later stages of the study, in line with the iterative nature of the analytical work. Chapters 4, 5, and 6 were drafted, revised, and refined based on supervisor feedback throughout the final months of the project.

The completed Gantt chart in Figure A.1 illustrates the actual execution of the dissertation. It complements the original planned Gantt chart and provides a transparent view of how the project unfolded in practice.

3. Completed Gantt Chart

The chart below reflects the actual execution of the dissertation, including iteration cycles, overlapping stages, refinement periods, and dissertation writing performed in parallel with analytical work.

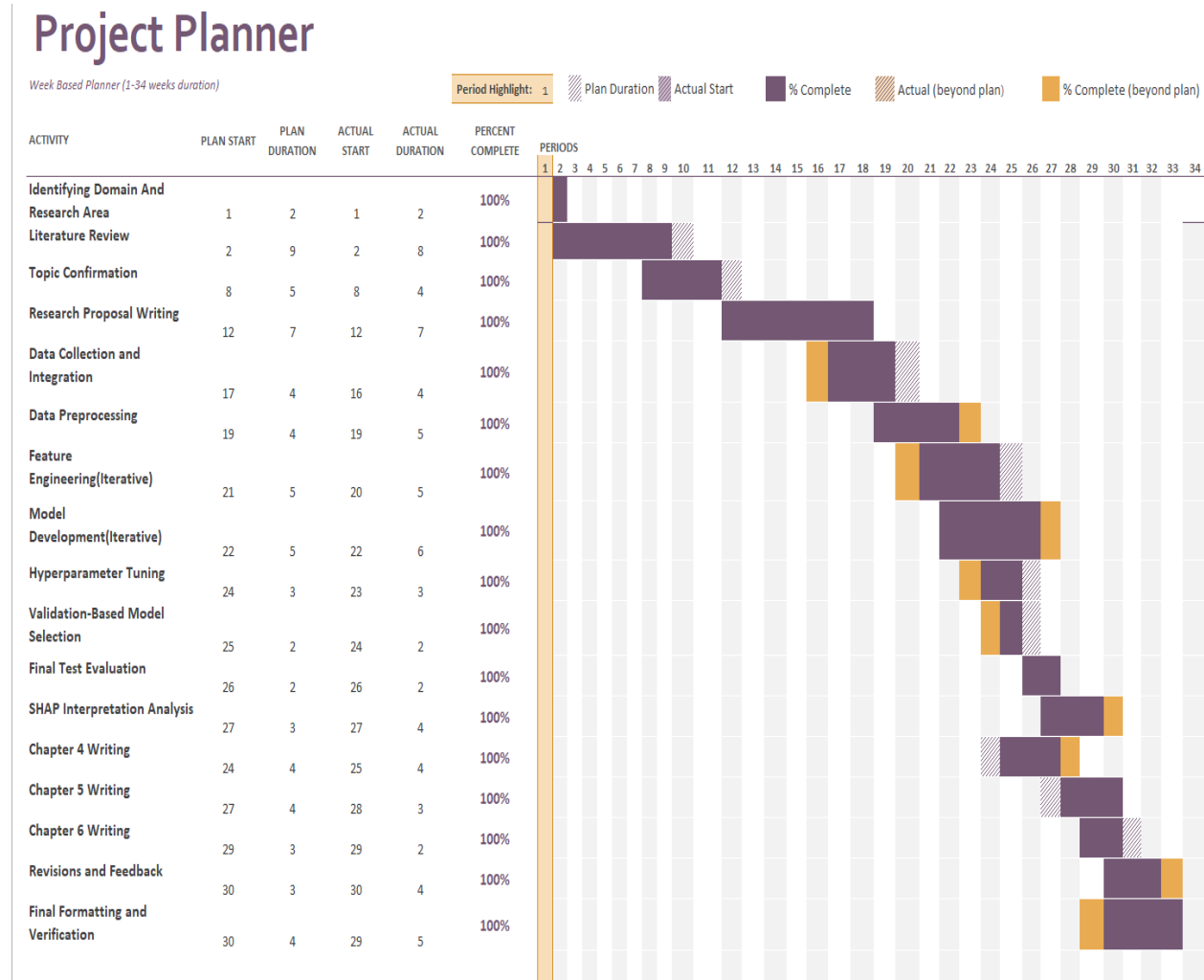


Figure A.1: Completed Weekly Gantt chart

Each cell represents one week. Where multiple stages overlap in the same week (for example, feature engineering iterations occurring alongside model development refinements), overlapping rows show extended durations that reflect the iterative nature of the analytical workflow.

4. Explanation of How Actual Execution Differed from the Planned Timeline

The completed weekly Gantt chart reflects the true progression of the dissertation, and differs from the original planned chart in the following ways.

4.1 Extended Literature Review Phase

The literature review extended beyond its original allocation to accommodate a deeper review of ablation-based studies, SHAP interpretability, macroeconomic integration, class imbalance handling, and construction cost prediction. This extension responded to the supervisor's feedback about scheduling ambiguity and provided a stronger analytical foundation.

4.2 Iterative Feature Engineering and Model Development

Feature engineering and model development were not linear activities. They required multiple iteration cycles due to preprocessing decisions, feature transformations, class-weight adjustments, ablation configuration testing, and repeated model comparisons. As recommended by the supervisor's feedback, more iteration weeks were included to allow refinement and stability.

4.3 Introduction of Additional Analytical Stages

Additional dedicated stages were incorporated for hyperparameter tuning, validation-based model selection, final test evaluation, and SHAP interpretation. These stages were introduced during the project to strengthen methodological rigor and align with distinction-level expectations.

4.4 Overlapping Analytical and Writing Activities

Dissertation writing progressed in parallel with the later analytical stages rather than as a final block. Chapter 4 was written while implementation was ongoing, Chapter 5 was written while results and interpretations were being finalized, and Chapter 6 was drafted after Chapter 5 was substantially complete. Revisions occurred throughout the final six weeks.

4.5. Expanded Contingency Buffers

The original proposal allocated only a single buffer week. The completed timeline reflects extended buffer periods surrounding feature engineering, model tuning, and SHAP interpretation to allow for iteration cycles. Additional buffer weeks were incorporated toward the end of the project for revisions and formatting.

6. Merged and Clarified Phases

Phases that were previously overlapping in the original chart (such as literature review and topic confirmation, and feature engineering and model development) were represented more accurately in the completed chart. These stages naturally merged during execution and are now presented in a clear, sequential and overlapping structure.

Together, these adjustments produce a completed timeline that responds directly to the supervisor's feedback, addresses the previous concerns about ambiguity, insufficient contingency, and unclear visualization, and provides a transparent and accurate representation of the actual dissertation workflow.